

Stephen Murphy

I help businesses deliver products to market through solid engineering design, ambitious goal-setting, effective communication between project teams, and positive work cultures.

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WORK EXPERIENCE

Sr. Staff Mechanical Engineer 05/2025 - Present
Shield AI San Diego, CA

- Architected and owned the conceptual development of the X-BAT wing fold system which includes a full 180° folding mechanism to achieve an unprecedented spotting factor of 0.33 on a 39 ft wingspan fighter jet.
- Designed the environmental cooling system (ECS) for the X-BAT Hopper prototype including heat exchangers, plumbing, manifolds, fluid storage, instrumentation, and air cooling.
- Delivered CAD and ASME Y14.5-2009 drawings using Siemens NX, FEA in Ansys and Simcenter, tolerance stacks, hand calcs, and work instructions.

Staff Mechanical Engineer 12/2023 - 04/2025
Shield AI San Diego, CA

- Owned the V-BAT wing from concept to production with novel designs for a cam-lock attachment mechanism, rotating pitot-static probe, rib-free composite structure, and various electronics and PCBA integrations.

Mechanical Design Manager 01/2023 - 11/2023
SkySafe San Diego, CA

- Led the redesign of two drone detection platforms from concept to production for improved ease of manufacture, cost, and reliability. This included thermal and structural FEA, CFD, CAD, drawings per ASME Y14.5-2018, RoHS & CE compliance reviews, prototyping, and testing.

Sr. Systems Engineer 09/2022 - 01/2023
Reverb Industrial San Diego, CA

- Architected and designed a novel ultrasonic metal 3D printer prototype.

Design Manager, AD & NPD 03/2021 - 09/2022
SharkNinja Boston, MA

- Owned the product design engineering of Shark FlexStyle, HyperAir, SpeedStyle, and SmoothStyle high volume consumer electronics products from PoC to mass production at volumes > 100k units/year.
- Led and performed mechanical design work including CAD (CREO), electronics integration, injection molding, DFM, DFA, CFD, FEA, DFMEA, statistical tolerance analysis, and design of experiments (DOE).
- Promoted to US Personal Care Engineering Lead where I led a team of 4 AD/NPD engineers and represented product development cross-functionally.

Engineering Manager, Mechanical Engineer 10/2015 - 01/2020
Intel (previously MAVinci until Intel acquisition in 2016) St. Leon-Rot, Germany

- Developed a commercial UAS from scratch including a 2 kg aircraft, batteries, GNSS base station, remote, and charger which resulted in my promotion to lead a team of 9 mechanical engineers.
- Designed and built a 3-axis CNC machine to reduce part lead times by 75%.
- Led a 9-person software team to the successful release of Intel Mission Control, a drone flight planning application for Windows, and personally programmed UIs in Java.

STRENGTHS

CAD

- Siemens NX
- Creo
- SOLIDWORKS
- Fusion 360

Software & Simulation

- MATLAB
- Python
- Ansys
- Java
- Simcenter 3D
- COMSOL

Mechanical Design

- Tolerance analysis
- GD&T
- Actuators
- Machining
- Composites
- DFM/DFA
- Acoustics
- Vibration
- Batteries
- Fluid systems
- Injection molding
- 3D Printing
- DFMEA
- Motors
- Thermals

Leadership

- Team building
- Architecture
- Cross-functional execution
- Engineering management
- Technical ownership

EDUCATION

Master of Engineering in Mechanical Engineering

University of Victoria 2020

Project: Thermo-electrochemical Model of a Zinc-air Flow Battery

Bachelor of Engineering in Mechanical Engineering with Co-op

University of Victoria 2015

Top 15% of class; honors thesis on high-altitude, long-endurance biplane design

LANGUAGES

English: Native speaker

German: Conversational

French: Limited